

Reverse Dual Photography

Adding the projector as a third camera for closed-loop 3D reconstruction

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June 8, 2026

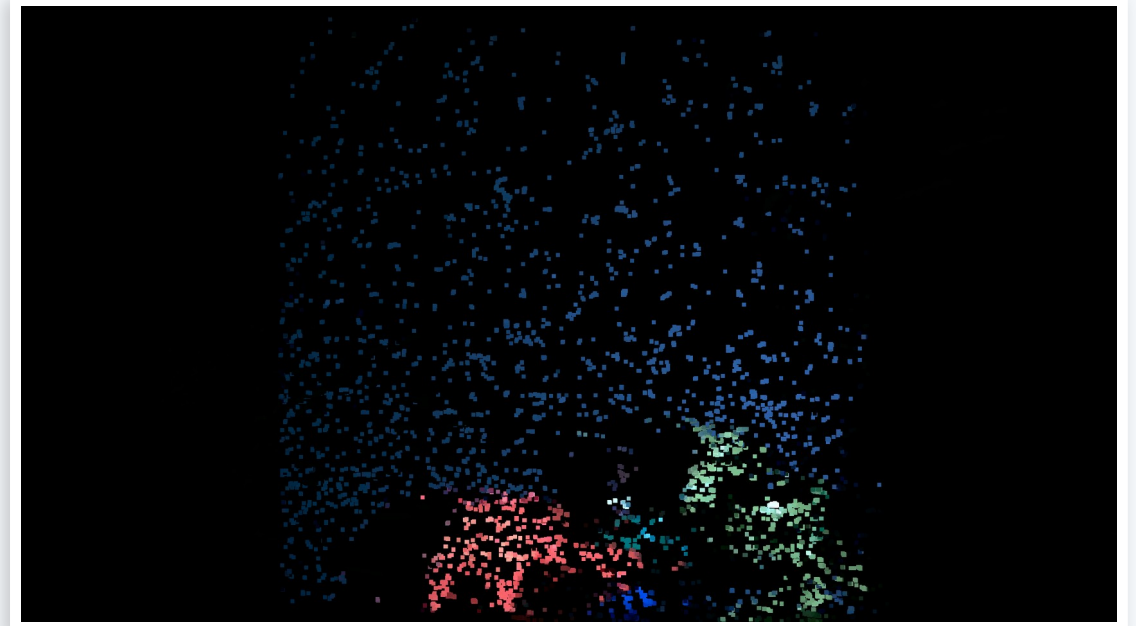
Dual Photography

Dual photography (Sen et al., 2005)

Measure the light transport between a projector and a camera with structured illumination, and you can re-render the scene from the projector's point of view — projector and camera swap roles.

Our system

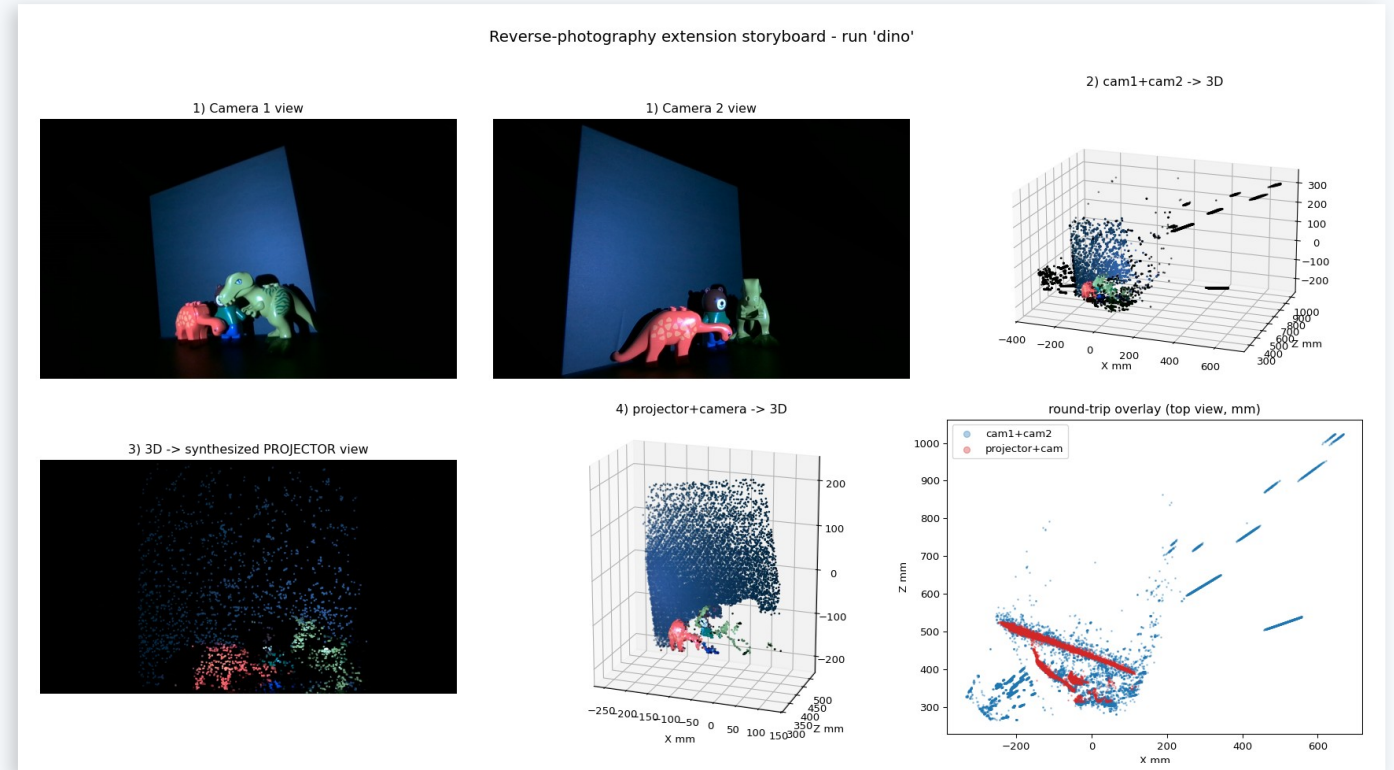
A structured-light scanner: two RealSense cameras + one projector. Every camera pixel is decoded to the projector cell that lit it — a dense projector ↔ camera correspondence.



Scene re-rendered from the projector's viewpoint (our result)

The projector as a third camera

- 1 Two cameras → 3D** Stereo-calibrate the cameras (checkerboard), triangulate the matched pixels into a metric point cloud.
- 2 Calibrate the projector** From those 3D points and their projector pixels, calibrate the projector as a 3rd camera — no extra capture.
- 3 Reverse: synthesize** Project the 3D back into the projector plane to rebuild the projector's 2D view.
- 4 Reconstruct again** Triangulate projector + one camera into an independent 3D cloud, and compare.



Full pipeline on a real object (dino): camera views → 3D → projector view → projector+camera 3D → overlay

RESULTS

The loop closes

~1 px

camera 3D reprojection error

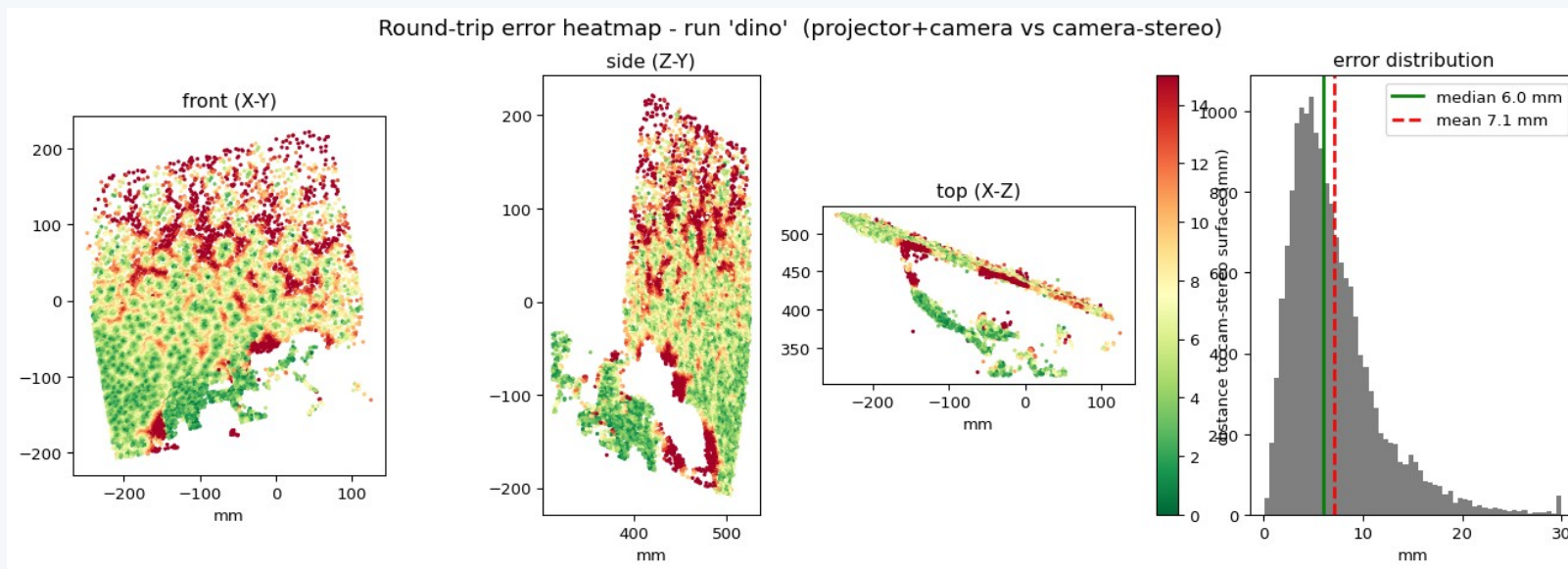
3.2 px

projector calibration RMS

~7 mm

round-trip agreement (median)

Two independent reconstructions of the same object — camera-stereo and projector+camera — agree to ~7 mm. Errors are local (red), concentrated at edges and occlusion boundaries; the surface is mostly green.



Limitations & next steps

Approximate projector intrinsics. A single shallow object under-constrains focal length (depth degeneracy), so we fixed square pixels and a centered principal point.

Dedicated projector calibration. Project a checkerboard onto a flat plane at several depths → tighter intrinsics → lower round-trip error.

Round-trip distance as a metric. The camera-stereo vs projector+camera agreement gives the system a built-in accuracy check.